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Development of Failure Diagnosis Method Based on Transient Information of Nuclear Power Plant

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This paper proposes a new method of failure diagnosis of nuclear power plant (NPP). Transient behavior of the NPP includes ample failure information even for a limited period of time from the failure onset. We tried to develop a diagnosis system with high capability of identifying the failure cause and of estimating failure severeness. The Walsh function transformation of transient time series data and the reduction of the Walsh coefficients into ternary valued amplitude indicators were utilized to extract the essential characteristics of failure. The correspondences of the transient characteristics and causes were summarized in a failure symptom database. A method of ternary tree search using an information measure as a heuristic strategy was adopted to conduct the efficient retrieval of failure causes in the database. Through numerical experiments using a simulation model of a NPP, the diagnostic capability of the system was proved to be satisfactory.

***KEYWORDS:** Bayes theorem, database, failure diagnosis, failure mode analysis, failure symptom, heuristic search, information measure, Kalman filter, nuclear power plants, reliability, transients, tree search, Walsh transformation*

I. INTRODUCTION

The identification of plant status by operators in abnormal situation is highly important for the enhancement of safety in nuclear power plants (NPPs). Within these several years, various kinds of failure diagnosis systems, to be used to support operator's decision making, have been developed. Typical examples of the techniques employed in the diagnosis systems are cause consequence tree (CCT)[1] and knowledge engineering[2]. Although these techniques gave successful results in small scale tests, several problems have to be solved for realistic applications. First, the number and complexity of CCTs and rules (diagnostic knowledge) tend to grow impractically large. The generation of CCTs and rules becomes a difficult, time consuming and expensive task. Also, the validity of the trees and rules becomes hard to assure.

The diagnostic systems using the aforementioned techniques utilize simplified descriptions of the dynamic behavior (*e.g.* temperature is increasing and pressure is decreasing.) of the objective plant. The dynamic behavior of the plant have to be treated in more detailed manner for a better performance of the diagnosis system. Although some techniques of utilizing the more detailed dynamic information of plant state were attempted[3][4], further developments are to be made to fully utilize the diagnostic information contained in the transient behavior.

From the viewpoints mentioned above, we propose following new techniques :

(1) Automated generation of diagnostic knowledge from transient behavior obtained by a simulator of the objective plant
(2) Efficient utilization of diagnostic informa-

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tion in transient behavior of the objective plant

(3) Design and implementation of diagnostic knowledgebase in a structured manner suitable for efficient information retrieval.

The generation of diagnostic knowledge by men makes the diagnosis system less reliable, because of the possibility of introducing the human-originated errors such as neglection and misunderstanding in the generation stage. The automated generation of diagnostic knowledge from transient behavior enhances the reliability of the diagnosis system on the condition that the transient behavior is obtained from a credible simulator of the objective plant. The credibility of the simulation was studied in our previous papers[5][6]. **Figure 1** shows the block diagram of the failure diagnosis system proposed in this report. The credible simulator derives each sensor output of the plant in time domain as numerical time series. The anomaly behavior in each sensor output caused by a failure is detected and segmented by utilizing the technique of Kalman filtering[7]. The segmented transient data are then transformed by Walsh function expansion[8] into Walsh coefficients. These coefficients are reduced into a set of ternary valued amplitude indicators using a posteriori probability obtained by Bayes theorem[9]. These indicators represent the failure symptom contained in the transient data. Each of the derived failure symptoms is related to its failure cause and severeness in the form of a ternary tree. These relationships, each having a ternary tree structure, are aggregated as the failure symptom database. The retrieval of failure causes and severeness is efficiently performed in this data structure. Based on the observed failure symptom from the objective plant, the diagnosis program retrieves failure causes and severeness in the failure symptom database using an information measure[10] as a heuristic strategy for efficient and reliable retrieval. This procedure of failure causes identification is conducted for each sensor output. The diagnostic result is obtained by selecting the AND set of all the candidate sets of the failure cause enumerated from all objective sensor. As the mutual relations among each sensor are considered in this final procedure, this diagnosis method has no problem like impractically large number of rules or CCTs.

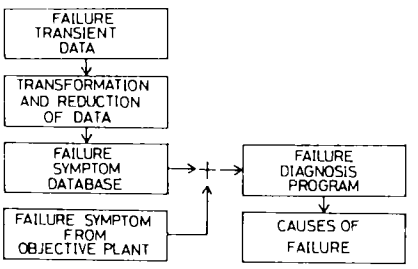


Fig. 1 Overall structure of failure diagnosis system

Chapter II describes the techniques of failure symptom database development. Chapter III shows the algorithm of identifying failure cause and severeness. The applicability of this system in practical use is demonstrated in Chap. IV.

II. Failure Symptom Database

Figure 2 explains the procedure of generating the failure symptom database. The transient behavior of the plant caused by the failure are obtained as numerical time series data by the simulation code with high credibility. First, the onset of the failure is detected in this time series, and the data span covering the initial phase of the abnormal behavior is segmented.

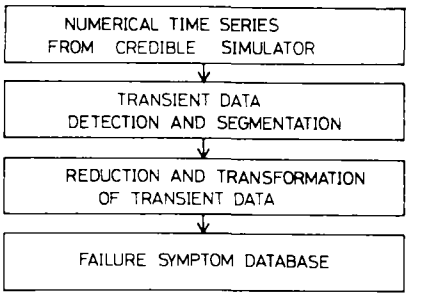


Fig. 2 Generation of failure symptom database

1. Detection and Segmentation of Transient Data

Following properties are required in the detection methodology of abnormal behavior in the numerical time series :

(1) Applicability to various initial, steady state, conditions

(2) Sufficiently high sensitivity without false alarm.

To satisfy these requirements, a method of anomaly detection was developed utilizing the technique of Kalman filtering[7].

We introduce following assumptions with signal and measurement processes :

$$x(k)=x(k-1)+u(k), \quad (1)$$

$$y(k)=x(k)+w(k). \quad (2)$$

Here, $x(k)$ and $y(k)$ are the process value and the measurement value of a sensor signal at time k, respectively; $u(k)$ and $w(k)$ are the process and the measurement noise, respectively. The noise processes are assumed to satisfy the conditions of stationarity and whiteness. Under these conditions, the least squares estimate $x_e(k)$ of $x(k)$ is derived by the Kalman filtering algorithm as follows :

$$p(k)=1/[1/\sigma_w^2+1/\{p(k-1)+\sigma_u^2\}], \quad (3)$$

$$x_e(k)=x_e(k-1)+\{p(k)/\sigma_w^2\}\{y(k)-x_e(k-1)\}. \quad (4)$$

Here, σ_u^2 and σ_w^2 are the mathematical expectations of variance of $u(k)$ and $w(k)$, respectively.

The detection of failure onset is performed as follows. First, prediction error $r(k)$ and mathematical expectation of error variance $E(r(k)^2)$ are defined as

$$r(k)=y(k)-x_e(k-1), \quad (5)$$

$$E(r(k)^2)=p(k-1)+\sigma_u^2+\sigma_w^2. \quad (6)$$

Utilizing the χ^2-test, the decision is made with 95% confidence level by the following rule :
if

$$r(k)^2<3.841E(r(k)^2), \quad (7)$$

then the process is normal, otherwise, *i.e.*

$$r(k)^2\geqq 3.841E(r(k)^2), \quad (8)$$

then the process is abnormal. The time series data after the first abnormal detection are regarded as the anomaly transient data. In actual applications, the threshold of $E(r(k)^2)$ might be defined larger than the value mentioned above to reduce the possibility of false alarm.

Following the detection of the failure onset, the signal records covering the major portion of the anomaly transient are segmented. After certain period of time, the feedback effects in the plant, the influence of secondary failures and the status change of the other equipment (*e.g.* relief valve open, auxiliary pump start, *etc.*) become dominating in the symptom. Consequently, the dynamic behavior can become too complicated to be analyzed by a simple algorithm. It was judged more reasonable to try to extract the diagnostic information from the early phase of the transient. From this reason, the transient behavior of the plant, certain period of time following the failure onset, was segmented to represent the anomaly characteristics. Various periods of the segmentation were examined in the diagnosis test in case of secondary loops of pressurized water reactor. The 60 s of time was selected as the suitable period, because the feedback effects in the plant appears after typically 60 s following the failure onset.

2. Transformation and Reduction of Transient Data

The detection and segmentation procedures provide us with primary failure symptoms still in the form of time series. However, the correspondence between the segmented numerical transient data and the failure cause cannot be established easily, because of the problems given below :

(1) The size of the numerical failure symptom database become impractically large.

(2) The search procedure of the true cause out of a large number of failure symptoms in the database becomes time-consuming. It is unrealistic to try to conduct the exact pattern matching between the observed and prestored failure symptoms in the form of time series data.

Following techniques of transformation and reduction of the segmented time series are introduced to overcome these difficulties. We paid our attention on the following point in this study. The key information for the diagnosis is the macroscopic variation patterns of the signals. The exact amplitude and timing of numerical time series seem to be less important, because experienced operators can actually

make correct decisions about the cause of the failure by observing the macroscopic behavior, rather than by quantitative details, of the transient. Bearing this thought in mind, it was judged reasonable to reduce the failure symptom into a modest number of feature parameters. We adopted an orthogonal function transformation as an efficient method of numerical data transformation without losing essential diagnostic information.

The selection of a suitable orthogonal function set is important issue of the transformation algorithm design. In this study, the Walsh function transformation[8] was adopted. This function system is a set of rectangular wave patterns, which is known to be effective in characterizing the digital time series data.

Typical observations, or descriptions, of the plant state, with which a human expert can conduct the task of failure cause identification, are given below :

(1) Signal 1 increased in a stepwise manner, kept the constant value for a while, then showed a ramp decrease.
(2) Signal 2 showed a ramp decrease until it reached a saturation level.
(3) Signal 3 showed a quick dip, then increased gradually.

The Walsh function system is more suitable than the other conventional continuous function systems, *e.g.* Fourier, Laguerre *etc.*, to transform these typical variation patterns. Each anomaly transient is transformed into a set of transformation parameters (Walsh coefficients), and labeled by the corresponding failure tag. The failure symptom database consists of the sets of these tags and parameters. The Walsh function transformation is terminated at a relatively low order (typically 15 in this study) to avoid unnecessary increase in the size of the database. It is believable that the higher order parameters do not include the information of the macroscopic structure of the failure symptom. The fast Walsh transformation (FWT)[8] algorithm was used as the actual procedure of the data conversion in an efficient manner.

Each feature parameter, corresponding to the transformation coefficient, is then reduced to define "amplitude indicator" for higher retrieval efficiency. For simplicity of explanation, let us focus on one particular signal among many observable signals. First, all the sample data of typical anomaly transients corresponding to various kinds of failure causes are transformed into feature parameters. We obtain a set of parameters for each transformation order i with these samples. Within this set, the parameter amplitude shows a certain empirical distribution. Each parameter set i is classified into three sectors with identical areas, corresponding to the ternary valued indicators, simply by finding two internal boundaries in the distribution. The parameters belonging to the sectors for high, middle and low amplitude values are labeled by the amplitude indicators "+", "0" and "−", respectively. A priori probabilities $P_i(+)$, $P_i(0)$ and $P_i(-)$ for a feature parameter in the set to be labeled by the indicators "+", "0" and "−", respectively, obviously satisfy the relation,

$$P_i(+)=P_i(0)=P_i(-)=1/3. \tag{9}$$

Furthermore, we can define a priori conditioned probability distributions of the feature parameter w_i of the i-th order by assuming a suitable amplitude distribution function of w_i in the sample set. The feature parameter w_i obtained from sample data is expected to have Gaussian distribution for the uncertainty of observation. We denote $P(w_i|+)$, $P(w_i|0)$ and $P(w_i|-)$ as a priori Gaussian probability distributions of w_i for the cases, when the indicators are "+", "0" and "−", respectively. This procedure is repeated for all the observed signals to be considered in diagnostic decision making.

A posteriori conditioned probabilities for the indicators "+", "0" and "−" on the condition that the observed value of the i-th order coefficient is w_i are obtained utilizing the Bayes theorem[9]. For example, a posteriori conditioned probability of the indicator "+" is calculated as follows :

$$P(+|w_i)=\frac{P(w_i|+)P_i(+)}{P(w_i|+)P_i(+)+P(w_i|0)P_i(0)+P(w_i|-)P_i(-)}. \tag{10}$$

Figure 3 shows one example of $P(+|w_i)$, $P(0|w_i)$ and $P(-|w_i)$. In this case, the largest a posteriori probability for an observed w_i is $P(0|w_i)=0.6$. Therefore, the amplitude indicator value for the w_i is judged to be "0".

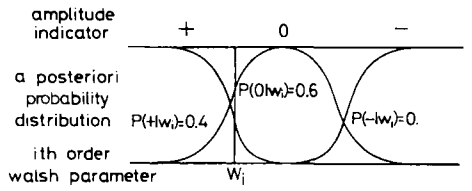


Fig. 3 Decision of i-th order parameter amplitude indicator by a posteriori probability distribution

3. Structure of Database

By the transformation and reduction described above, the computational time for failure cause diagnosis was significantly reduced. In spite of this time reduction, however, the task of diagnosis still required considerable computational time for examining a large number of failure symptoms. For further reduction of the computational time, the database was designed to have a specific tree structure as illustrated in **Fig. 4.**

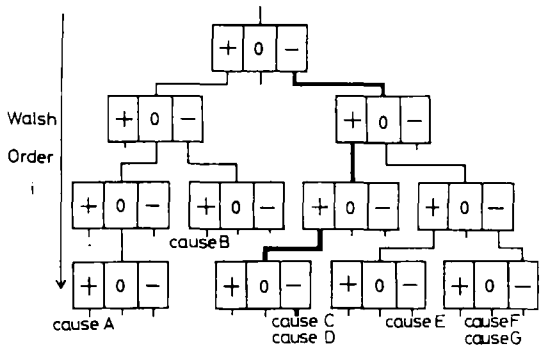


Fig. 4 Correspondence of symptoms and causes in ternary tree structure

Each node has three branches labeled "+", "0" and "−", corresponding to three values of the amplitude indicator. The nodes at a higher layer correspond to lower transformation order and *vice versa.* For example, the indicator set [−++−], linked by the thick lines, corresponds to the causes C and D. Since the indicator set consists of four elements in this example, only four comparisons are needed to determine the failure cause. This structure of the database allows us a highly efficient retrieval. When new data are obtained, implementation of the data to the database is performed easily using the recursive algorithm, as far as the amount of the data is small. For a large amount of new data, however, the redefinition of a priori conditioned probability distribution would be desirable.

Each sensor output can show specific behavior for each failure. Therefore the tree shown in Fig. 4 was generated for each sensor.

III. Failure Cause Identification

The procedure of failure cause identification developed in this study consists of two steps. In the first step, candidates of failure cause are retrieved based on each sensor output. In the second step, the most likely candidates are determined among the candidates obtained at the first step.

1. Search Procedure Based on Information Measure

This section describes the determination procedure of failure cause candidates based on the database obtained for single sensor output. The observed anomaly transient in the sensor output is transformed and reduced into a set of amplitude indicators by the procedure explained in Chap. II. The failure cause retrieval is performed by the search procedure utilizing an information measure[10], which is defined by the (−1) times logarithm of a posteriori probability of the indicator. This measure was introduced to overcome a specific difficulty, *i.e.* reasonable decision making based on uncertain knowledge. It is quite plausible that the two transients with slightly different failure symptoms are obtained by the causes to be identified as "same", owing to a small difference in magnitude and/or time characteristics of the failure. In other words, the indicator sets corresponding to the same failure cause should be regarded as identical despite the small difference. At the same time, the algorithm is requested to distinguish two failure causes, which are related to apparently resembling symptoms. To satisfy these conflicting requirements, the search within the database is performed by successive evaluation of the average information measure at each search step.

The retrieval algorithm based on the observed set of amplitude indicators is summarized as following recursive procedure :

(1) Let $i=0$, $E(im)=0$.
(2) Call retrieval subroutine.
(3) End.
(4) Retrieval subroutine: let $i=i+1$.
(5) Calculate $im(+)$, $im(0)$ and $im(-)$.
(6) Let $E(im)'=\{(i-1)/i\}E(im)+\{1/i\}im(+)$.
(7) If $E(im)'<\varepsilon$ and the branch labeled "+" is connected with $(i+1)$-th layer node, then let $E(im)=E(im)'$ and call retrieval subroutine.
(8) Let $E(im)'=\{(i-1)/i\}E(im)+\{1/i\}im(0)$.
(9) If $E(im)'<\varepsilon$ and the branch labeled "0" is connected with $(i+1)$-th layer node, then let $E(im)=E(im)'$ and call retrieval subroutine.
(10) Let $E(im)'=\{(i-1)/i\}E(im)+\{1/i\}im(-)$.
(11) If $E(im)'<\varepsilon$ and the branch labeled "−" is connected with $(i+1)$-th layer node, then let $E(im)=E(im)'$ and call retrieval subroutine.
(12) If $i=N$ then failure cause tags are listed up as candidates.
(13) Return to step $(j+1)$ when called at step j.

Here, i is the depth index of a node in the tree structure, *i.e.* the order of the Walsh function transformation. The symbol N is the maximum depth index, *i.e.* the maximum order, $E(im)$ the average information measure, $im(+)$, $im(0)$ and $im(-)$ information measures of each indicator value, given by $-\log P(+|w_i)$, $-\log P(0|w_i)$ and $-\log P(-|w_i)$, respectively. The parameter ε is a threshold value of average information measure for the determination of the branches to retrieve. Since the amplitude indicator can be determined uniquely when its conditional probability is greater than 0.5 (*i.e.* when the average information measure becomes smaller than $-\log 0.5$), the value of ε was selected to be $-\log 0.5$.

The decision by the threshold value ε defines a heuristic rule. This heuristic rule makes the retrieval highly flexible. Note that the average information measure of each failure cause candidate become a measure of the certainty of the result. The failure candidate identified with a small value of the information measure is regarded to be highly probable. The minimum value of 0.0 is obtained when the result is identified with 100% certainty.

2. Decision of Failure Cause Based on Multiple Sensor Signals

As various sensor signals are obtained from the plant, candidates of the failure cause are enumerated for each signal. When the enumerated candidates from each signal are mutually consistent, the determination of the failure cause is straightforward. When the candidate sets are not perfectly identical, however, some additional guidelines are needed for the decision making. The most conservative guideline is to select the AND (*i.e.* commonly obtained) set of all the candidate sets, as shown in **Fig. 5.**

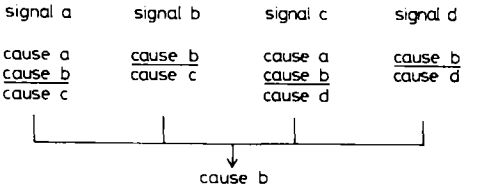


Fig. 5 Decision of causes for set of signals

This guideline is most appropriate for the purpose of minimizing the false alarm probability. An alternative option is to apply the majority voting rule which is less conservative than the aforementioned one. We adopted the conservative guideline in the present study. The majority voting might be adopted when smaller probability of miss alarm is required.

IV. Results and Discussion

1. Generation of Failure Symptom Database

As the first step of verifying the effectiveness of this failure diagnosis technique, the failure symptom database was generated based on a set of transient data obtained from a conventional numerical simulator of the secondary loop of a PWR[11], and the failure cause identification was attempted for the same set of test data. Typical examples of the transients caused by the stepwise decrease in three element controller (TEC) output and by saturation of feed water flow sensor output, are illustrated in **Figs. 6** and **7**, respectively. A time step length in these figures is equal to two

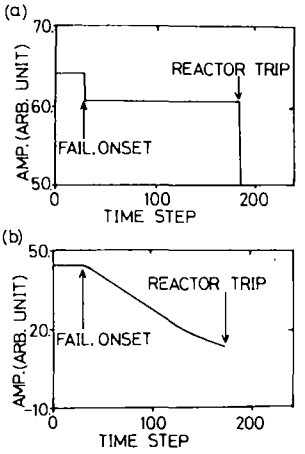


Fig. 6 Responses of TEC output (a) and water level in SG (b) for step decrease of three element controller output

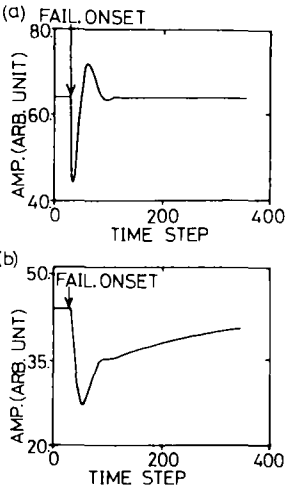


Fig. 7 Responses of TEC output (a) and water level in SG (b) for saturation of feed water flow sensor output

seconds period. The time dependent variations of the signals are relatively simple in the first example, while they become more complicated in the second one. Five signals considered in this numerical experiment are shown in **Table 1**.

Table 1 Signals utilized in experiment

Signal 1	Steam generator (SG) water level
2	Three element controller (TEC) output
3	Steam generator (SG) supply water flow
4	Main steam line pressure
5	Main steam line flow

Tables 2 and **3** summarize the result of diagnosis obtained for the transient shown in Figs. 6 and 7, respectively. Analysis of Signals 1~3 resulted to the unique indication of the stepwise decrease of TEC output. On the other hand, failure causes obtained by analyzing Signals 4 and 5 are not unique. These signals enumerated the stepwise decrease of TEC output and the leakage of SG heat exchange pipe. The failure symptoms of these failures appeared in the main steam line pressure (Signal 4) are identical, because each failure keeps the pressure value unperturbed. The main steam line flow (Signal 5) decreases, because of the decrease of the supply water flow to SG in each case. The decrease of TEC output directly decreases the supply water flow. On the other hand, the leaked water from heat exchange pipe increases the SG water level. This effect forces the TEC output, and, resultantly, the supply water flow to be low. These effects make the symptoms of the two failures mutually resembling. For this reason, analysis of Signals 4 and 5 failed to distinguish the two failure causes. The values of the information measure become larger for these signals, implying that the results have lower certainty. By taking the AND of all candidate sets, however, the true failure cause was determined uniquely.

In the case shown in Table 3 obtained for the saturation of feed water flow sensor output, exactly identical diagnostic results are obtained from each signal, suggesting that the result is highly plausible. The values of information measure also indicate the certainty of the result. The correct diagnostic results were obtained for all the other tested transients of total 10 cases. The present diagnosis algorithm provides us with reasonable results as far as experienced transient data are concerned.

2. Diagnosis of Realistic Test Data

As the second step of verifying applicability of this system, the failure diagnosis was performed with the more realistic, unexperienced transients, which were generated by modifying the original transient data. In the example shown in **Fig. 8**, the test data were obtained

Table 2 Result of failure diagnosis of transient data "Step decrease of TEC output"

(a)

Signal No.	Parameter amplitude indicator
1	−+−−−0−−0000−0−−
2	−0000000000000−000
3	−000−−−0−−−+−−−+
4	0000000000000000000
5	0000000+0−−+000+

(b)

Signal No.	Estimated cause	Information measure
1	Step decrease of TEC output	0.09
2	*ditto*	0.11
3	*ditto*	0.14
4	SG heat exchange pipe leak from primary loop to secondary loop 12 t/h	0.21
	SG heat exchange pipe leak from primary loop to secondary loop 60 t/h	0.21
	Step decrease of TEC output	0.21
5	Step decrease TEC output	0.35
	SG heat exchange pipe leak from primary loop to secondary loop 12 t/h	0.58
Overall judgement	Step decrease of TEC output	0.18

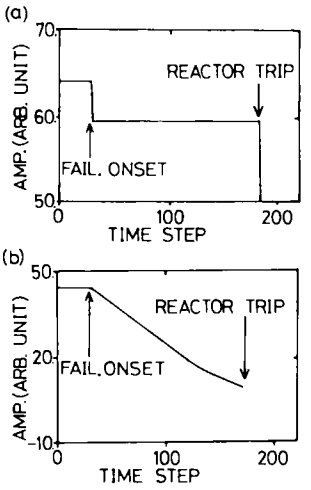


Fig. 8 Responses of TEC output (a) and water level in SG (b) for step decrease of three element controller output (with amplified magnitude by a factor of 1.2)

Table 3 Result of failure diagnosis of transient data "Saturation of feedwater flow sensor"

(a)

Signal No.	Parameter amplitude indicator
1	−+−+−+−+−−++−+−
2	−+−+++++++++−++++
3	−−+++++++++−++++
4	+0−0−−−0−−0+−−−0
5	++−−−−−−−−−+−−−+

(b)

Signal No.	Estimated cause	Information measure
1	Saturation of feedwater flow sensor	0.08
2	*ditto*	0.01
3	*ditto*	0.00
4	*ditto*	0.25
5	*ditto*	0.10
Overall judgement	*ditto*	0.09

by artificially amplifying the magnitude of the experienced signals by a factor of 1.2. Here, we implicitly assumed the linearity of transient response. The second example illustrated in **Fig. 9** was obtained by adding white noise having the amplitude of 10% peak to peak

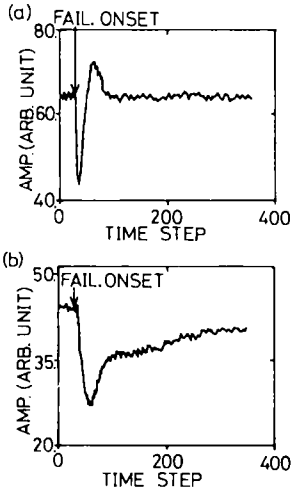


Fig. 9 Responses of TEC output (a) and water level in SG (b) for saturation of feed water flow sensor output (added white noise having the amplitude of 10% peak to peak variation of the undistorted signal)

variation of the undistorted signals. Note that the other test data distorted by white noise with a larger amplitude were also examined. Because of the amplitude modulation or the noise contamination, the test signals in this experiment are regarded to be unexperienced and different from the transient data used for database generation.

Tables 4 and **5** show the diagnosis results obtained for the realistic test data. In Table 4, Signal 4 gave the multiple causes, although the element of the AND set is unique. In Table 5, each signal show the identical result. In case of the data contaminated by white noise having the amplitude of 20% peak to peak variation, each signal could not show the common result. This is unavoidable since the substantial increase in the measurement noise should be attributed to another failure cause. Expansion of the database should be performed to cover significant increasing measurement noise. In the cases of Tables 4 and 5, the sets

Table 4 Result of failure diagnosis of transient data "Step decrease of TEC output" (amplified variation magnitude by a factor 1.2)

(a)

Signal No.	Parameter amplitude indicator
1	−+−−−0−−0000−0−−
2	−0000000000000−000
3	−+00−−−0−−−+−−−+
4	0000000000000000+
5	000+000+0−−+0−0+

(b)

Signal No.	Estimated cause	Information measure
1	Step decrease of TEC output	0.11
2	*ditto*	0.11
3	*ditto*	0.26
4	SG heat exchange pipe leak from primary loop to secondary loop 12 t/h	0.32
	SG heat exchange pipe leak from primary loop to secondary loop 60 t/h	0.32
	Step decrease of TEC output	0.32
5	Step decrease of TEC output	0.40
Overall judgement	*ditto*	0.24

Table 5 Result of failure diagnosis of transient data "Saturation of feed-water flow sensor" (added white noise having the amplitude of 10% peak to peak variation of the sensor signal)

(a)

Signal No.	Parameter amplitude indicator
1	−++−+−+−−−++−+−
2	−−−+++++++++−+++
3	−−+++++++++−+++
4	+0−0−−−0−−0+−−−0
5	++−−−−−−−−−+−−−+

(b)

Signal No.	Estimated cause	Information measure
1	Saturation of feed-water flow sensor	0.16
2	*ditto*	0.01
3	*ditto*	0.00
4	*ditto*	0.23
5	*ditto*	0.06
Overall judgement	*ditto*	0.09

of amplitude indicators are slightly different from those in Tables 2 and 3. The values of information measures are larger than those of Tables 2 and 3. These facts show the applicability of this diagnosis technique, although larger uncertainty was inevitable in the results.

Further study has been made to examine the capability of the present method to distinguish magnitude of failures. As the matter of fact, expansion of the symptom database to include failures with different magnitudes was necessary. Typical results obtained for faulty leak from the relief valve obtained from a conventional numerical simulator are shown in **Table 6**. In this example, the threshold ε is deliberately made greater than $-\log 0.5$ to evaluate the values of information measure corresponding to 2 and 6% leak. As is clearly shown by the values of information measure, the failure cause (6% leak) having the maximal similarity to the introduced failure (8% leak) was successfully identified. Through these numerical experiments, it was confirmed that the present method can be used not only for identifying failure cause but also for distinguishing failure magnitude.

Table 6 Result of distinguishing failure magnitude "8% leak from relief valve at main steam line"

Signal No.	Estimated cause	Information measure
1	6% leak from relief valve	0.46
	2% *ditto*	1.25
2	6% *ditto*	0.92
	2% *ditto*	1.43
3	6% *ditto*	0.54
	2% *ditto*	1.48
4	6% *ditto*	0.04
	2% *ditto*	0.04
5	6% *ditto*	0.44
	2% *ditto*	1.76
Overall judgement	6% *ditto*	0.48
	2% *ditto*	1.19

Note that 2% and 6% leak data are stored in the failure symptom database.

The performance of this algorithm obviously depends on the reliability and sufficiency of the test data utilized for generating the database. As one natural approach, we have been trying to develop a high performance simulator[5][6] which can reproduce the essential features of dynamic responses of the plant for various causes. It is also possible to collect experimental data obtained at a specified plant, and to use them as supplementary data for database generation. The issue of test data generation will be further pursued in our future study.

This diagnosis program was developed on the ACOS-1000 computer by the C language to take the advantage of its excellent flexibility, portability and maintainability. Only about 0.5 s was required as the computational time for diagnosing a failure. The size of the memories needed for this program was less than 20 K bytes. This fact implies that this algorithm is executable even by a mini-computer.

V. CONCLUSION

In this study, a new technique of failure diagnosis was introduced. The performance of the developed algorithm was evaluated through numerical experiments, resulting to the observations given below :

(1) The detection, segmentation, Walsh transformation and ternary reduction procedures were effective in storing the failure symptom database in reasonable memory size.

(2) The structural design of the database, and the search algorithm using the information measure as well, facilitated retrieval of failure causes within reasonable computation time.

(3) The credibility of diagnostic results was found to be high not only for experienced but for unexperienced (*i.e.* distorted) test data.

These results clearly show that the present technique has high potentiality as a tool for dependable diagnosis of failure-induced transients in real NPPs. Further improvement in the diagnostic capability can be attained by supplying larger amount of reliable test data for database generation.

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